

# WRF-Development Process

05 July 2023 15:12

1. Input config:  
Namelist.wps: [https://www2.mmm.ucar.edu/wrf/users/namelist\\_best\\_prac\\_wps.html](https://www2.mmm.ucar.edu/wrf/users/namelist_best_prac_wps.html)  
Namelist.input: [https://www2.mmm.ucar.edu/wrf/users/namelist\\_best\\_prac\\_wrf.html](https://www2.mmm.ucar.edu/wrf/users/namelist_best_prac_wrf.html)  
WRF-CHEM: [https://ruc.noaa.gov/wrf/wrf-chem/Users\\_guide.pdf](https://ruc.noaa.gov/wrf/wrf-chem/Users_guide.pdf)
2. WPS/Geogrid.exe: generate geo\_em.dxx.nc

1. **Download** the terrestrial input data
2. Edit the **&share** and **&geogrid** sections of the **namelist.wps** file for your particular domain set-up.



Run plotgrids.ncl to ensure your domain is in the right location before running geogrid.exe

```
ncl util/plotgrids.ncl
```

3. Run **geogrid.exe** (output is in the format of 'geo\_em.dxx.nc' - one file for each domain)

```
./geogrid.exe
```

3. WPS/Ungrib.exe: generate [FILE:YYYY-MM-DD\\_hh](#)

2. **Familiarize yourself with the data**

3. **Link** (with the UNIX command *ln*) the correct Vtable

-For example, if you are using GFS data, type:

```
ln -sf ungrib/Variable_Tables/Vtable.GFS Vtable
```

4. **Link** (with supplied script [link\\_grib.csh](#)) the input GRIB data

```
./link_grib.csh path_to_data
```

5. Edit the **&share** and **&ungrib** sections of the **namelist.wps** file. You only need to pay attention to the following parameters:  
start\_date ; end\_date ; interval\_seconds ; prefix

**Note:** Normally one will leave the "prefix" set to "FILE", except in cases where this may overwrite data.

6. Run ungrib.exe (output will be intermediate files in the format of FILE:YYYY-MM-DD\_hh - one file for each time)

```
./ungrib.exe
```

7. **Familiarize yourself with the intermediate files**

4. WPS/Metgrid.exe: generate met\_em.d0X.YYYY-MM-DD\_hh:00:00.nc

1. Edit the **&share** and **&metgrid** sections of the namelist.wps file.

2. Input to METGRID is the **geo\_em.dxx.nc** output files from GEOGRID, and the intermediate output files from UNGRIB (e.g., FILE:YYYY-MM-DD\_hh).

3. Run **metgrid.exe**

```
./metgrid.exe
```

Output from this program will be:

- met\_em.d0X.YYYY-MM-DD\_hh:00:00.nc - one file for per time, for each domain ("d0X" represents the domain).

5. WRF/test/em\_real

1. **Move** to the directory in which you plan to run the code (either **WRF/test/em\_real** or **WRF/run**).

-Both the **run/** and **test/em\_real/** directories will have all the files you need linked in, so it does not matter which one you choose to run the code in.

2. **Link or copy** (with the UNIX command *ln* or *cp*) the **met\_em** files to this directory.

```
ln -sf path_to_met_em_files/met_em.d0*
```

3. Edit the namelist.input file for your particular run. For descriptions of the namelist parameters, as well as suggestions for best practices, see our [Best Practices WRF Namelist](#) page.

4. **Run real.exe** (*verify that the program runs correctly*)

-You should have the following **output files** (default setup): **wrfinput\_d01** & **wrfbdy\_d01**

-This is true for single domain and default nested runs.

-If you plan to use a nested domain, you will have a **wrfinput\_dxx** file for each domain.

(more on this in 'nested case studies').

5. **Run wrf.exe** (*verify that the program runs correctly*)

-You should have the following **output files** (default setup): **wrfout\_dxx\_[initial\_date]** (one for each domain)

-Each file (by default) will contain all the forecast output times.

6. Use a [post-processing tool](#) to view the output.

Tutorial:

<https://www2.mmm.ucar.edu/wrf/OnLineTutorial/CASES/index.php>